

Build Your Advanced Agent With n8n

AI-Driven Multi-Agent Healthcare System

1. Introduction & Motivation

The integration of Generative AI (GenAI) and Large Language Models (LLMs) has significantly transformed the healthcare sector by enabling advanced natural language processing, improving access to medical information, and streamlining patient interactions. AI-powered solutions can assist in diagnosing conditions, providing health recommendations, and offering 24/7 support to patients.

PRE-REQUISITES: GET READY FOR THE SESSION

Before the session, complete the following:

Accounts Required:

- n8n Cloud or Self-hosted instance
- OpenAI or any LLM API key (with GPT-4 access)
- Airtable or Google Sheets account (for hospital/lab data storage)
- Twilio or similar SMS service for messaging (API key & verified number)

Basic Tools Setup:

- Ensure all APIs are working and whitelisted
- Ensure JSON agent definitions are uploaded into [n8n](#)
- Confirm you can test webhooks locally or via a live URL

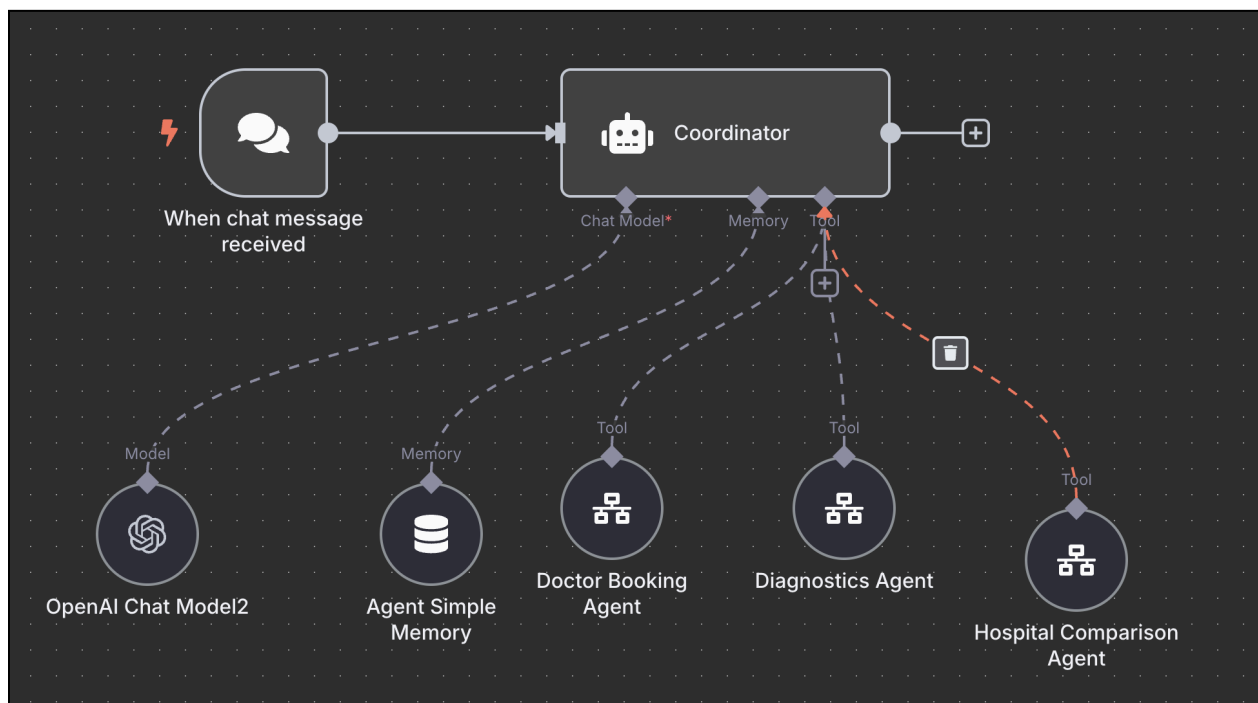
Local Files/Folders:

- Keep the following files ready (shared with you):
 - `Coordinator_Agent.json`
 - `Doctor_Booking_Agent.json`
 - `Hospital_Comparison_Agent.json`
 - `Diagnostics_Agent.json`

⚙️ Agent System Architecture

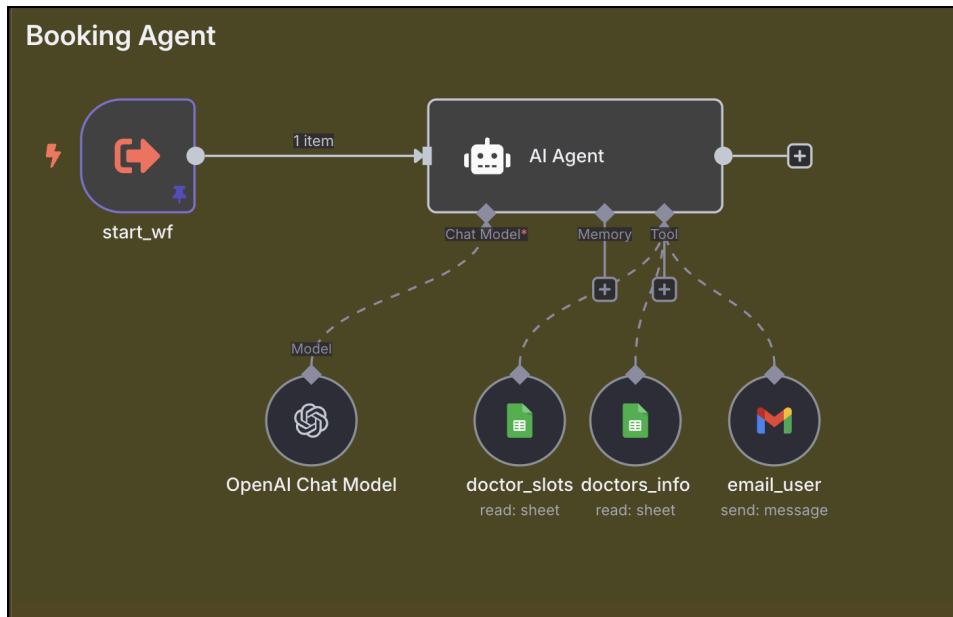
Coordinator Agent

- Parses the query
- Routes to relevant sub-agent (doctor, hospital, diagnostics)



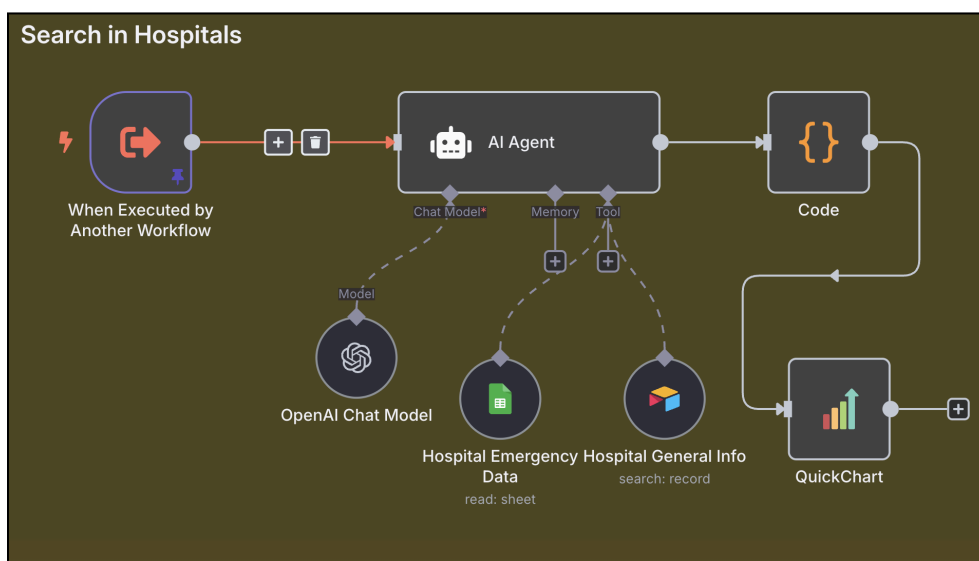
Doctor Booking Agent

- Finds slots from Google Sheets
- Suggests booking & confirms via email



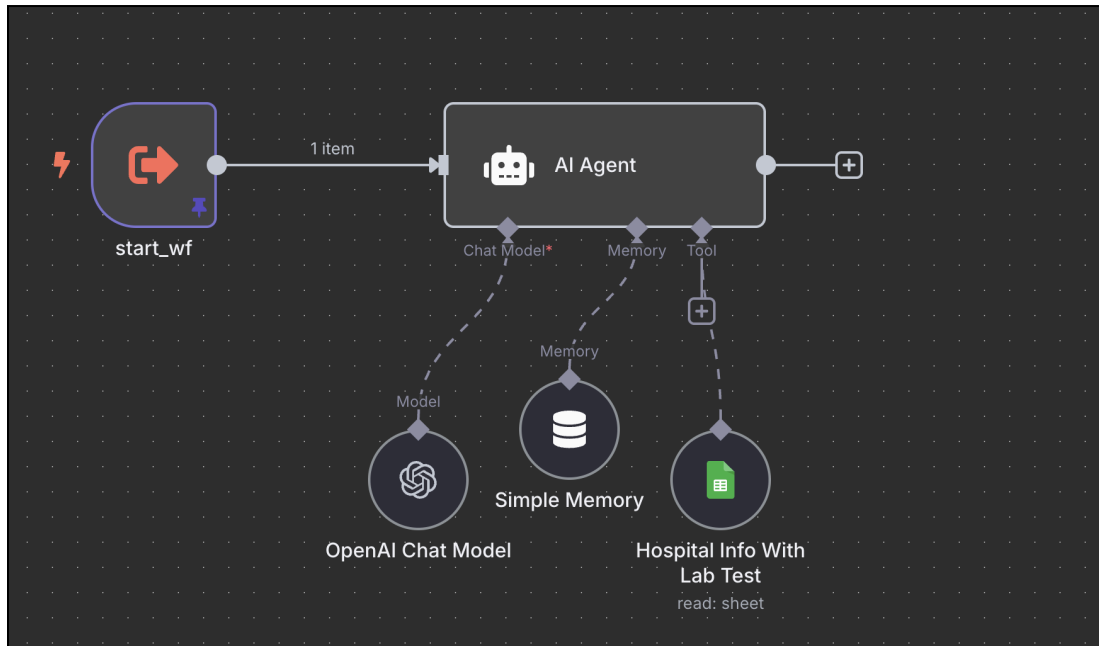
Hospital Comparison Agent

- Filters hospitals by city/specialisation/emergency availability
- Returns ranked list with contact info



Diagnostics Agent

- Maps symptoms to recommended lab tests
- Fetches test descriptions & prep steps from Google Sheets



Setup Steps

Step 1: Load JSON Agent Workflows

Upload all **.json** agent files into your **n8n** instance via the workflow import feature.

Step 2: Prepare Your Data Sources

- Create Google Sheets or Airtable bases for:
 - Doctor availability
 - Hospital database
 - Diagnostics test descriptions

Step 3: Connect APIs

- Link your OpenAI, Google Sheets/Airtable, and Twilio credentials in **n8n**
- Test them using **Credentials** in n8n UI

Step 4: Set Up Webhooks (Optional)

- Configure webhooks for incoming user queries if applicable (e.g., from a web app or chatbot)

Step 5: Test Each Agent Individually

- Run queries like:
 - "Show slots for Dr. Lee."
 - "Any ambulance available at 94404?"
 - "What would be the preparation instructions for cancer screening?"

Step 6: Combine Into Unified Flow

- Use the **Coordinator Agent** to route queries to other agents using conditional logic or LLM routing.
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Future Enhancements

- Add voice interface using Whisper + Twilio
 - Implement health record integration (FHIR)
 - Add multi-language response capabilities
 - Use LangChain agents for more complex orchestration
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Common FAQs

Q1: Can I use ChatGPT or Gemini instead of OpenAI GPT-4 API?

Yes, but ensure API supports tool-calling & function schemas.

Q2: What if I don't have Twilio?

You can use alternate SMS services (like Gupshup or MSG91) by customising the API node.

Q3: Will the workflows work if I only use Google Sheets (not Airtable)?

Yes. Update references accordingly within each agent workflow.
